통계열역학 (2012년도 1학기)

2차 과제 정답 (**총 65 점**)

1. **(총 20점)**

(a) **(5점)**

$$\begin{aligned} q_{elec}(T) &= \sum_{i=1} g_{ei} e^{-\beta \epsilon_{ei}} = \sum_{i=1} g_{ei} \exp\left[-\frac{hc\bar{\nu}}{k_B T} \right] \\ &= \sum_{i=1} g_{ei} \exp\left[-\frac{(6.62608 \times 10^{-34}\ \text{J s})(2.99792458 \times 10^{10}\ \text{cm s}^{-1})}{(1.38066 \times 10^{-23}\ \text{J K}^{-1})} \cdot \frac{\bar{\nu}(\text{cm}^{-1})}{T} \right] \\ &= \sum_{i=1} g_{ei} \exp\left[-(1.439\ \text{cm K}) \frac{\bar{\nu}(\text{cm}^{-1})}{T} \right]. \end{aligned} \tag{1}$$

위 식에서 g_{ei} 는 각 에너지 준위에서 축퇴도이고 ϵ_{ei} 는 i 번째 에너지 준위의 에너지이다.

$$q(150\ \text{K}) = 2 + 2e^{-(1.439\ \text{cm K})(121.1\ \text{cm}^{-1})/(150\ \text{K})} = 2.63$$

$$q(300\ \text{K}) = 2 + 2e^{-(1.439\ \text{cm K})(121.1\ \text{cm}^{-1})/(300\ \text{K})} = 3.12$$

(b) **(5점)**

$$\begin{aligned} \bar{S}(T) &= k_B \ln Q + \frac{\langle E \rangle}{T} \\ &= k_B \ln Q + k_B T \left(\frac{\partial \ln Q}{\partial T} \right)_{N_A, V} \\ &= k_B \ln q^{N_A} + k_B T \left(\frac{\partial \ln q^{N_A}}{\partial T} \right)_{N_A, V} \\ &= N_A k_B \ln q + N_A k_B T \left(\frac{\partial \ln q}{\partial T} \right)_{N_A, V} \\ &= N_A k_B \ln \left(2 + 2e^{-(1.439\ \text{cm K}) \frac{\bar{\nu}(\text{cm}^{-1})}{T}} \right) + N_A k_B T \left(\frac{\partial \ln \left(2 + 2e^{-(1.439\ \text{cm K}) \bar{\nu}(\text{cm}^{-1})/T} \right)}{\partial T} \right)_{N_A, V} \\ &= R \ln \left(2 + 2e^{-(1.439\ \text{cm K}) \bar{\nu}(\text{cm}^{-1})/T} \right) + RT \left(\frac{\partial \ln \left(2 + 2e^{-(1.439\ \text{cm K}) \bar{\nu}(\text{cm}^{-1})/T} \right)}{\partial T} \right)_{N_A, V} \\ &= R \ln \left(2 + 2e^{-(1.439\ \text{cm K}) \bar{\nu}(\text{cm}^{-1})/T} \right) + RT \frac{2e^{-(1.439\ \text{cm K}) \bar{\nu}(\text{cm}^{-1})/T}}{2 + 2e^{-(1.439\ \text{cm K}) \bar{\nu}(\text{cm}^{-1})/T}} \frac{(1.439\ \text{cm K}) \bar{\nu}(\text{cm}^{-1})}{T^2} \\ &= R \ln \left(2 + 2e^{-(1.439\ \text{cm K}) \bar{\nu}(\text{cm}^{-1})/T} \right) + \frac{(1.439\ \text{cm K}) \bar{\nu}(\text{cm}^{-1}) e^{-(1.439\ \text{cm K}) \bar{\nu}(\text{cm}^{-1})/T}}{1 + e^{-(1.439\ \text{cm K}) \bar{\nu}(\text{cm}^{-1})/T}} \frac{R}{T} \\ &= R \ln \left(2 + 2e^{-(1.439\ \text{cm K}) \bar{\nu}(\text{cm}^{-1})/T} \right) + \frac{(1.439\ \text{cm K}) \bar{\nu}(\text{cm}^{-1})}{e^{(1.439\ \text{cm K}) \bar{\nu}(\text{cm}^{-1})/T} + 1} \frac{R}{T} \end{aligned} \tag{2}$$

(2) 식으로부터

$$\begin{aligned}\bar{S}(150\text{ K}) &= (8.31451\text{ J K}^{-1}\text{ mol}^{-1})\ln\left(2+2e^{-(1.439\text{ cm K})(121.1\text{ cm}^{-1})/(150\text{ K})}\right)\\ &\quad+\frac{(1.439\text{ cm K})(121.1\text{cm}^{-1})}{e^{(1.439\text{ cm K})(121.1\text{cm}^{-1})/(150\text{ K})}+1}\frac{(8.31451\text{ J K}^{-1}\text{ mol}^{-1})}{(150\text{ K})}\\ &= \mathbf{10.3}\text{ J K}^{-1}\text{ mol}^{-1}\end{aligned}$$

$$\begin{aligned}\bar{S}(300\text{ K}) &= (8.31451\text{ J K}^{-1}\text{ mol}^{-1})\ln\left(2+2e^{-(1.439\text{ cm K})(121.1\text{ cm}^{-1})/(300\text{ K})}\right)\\ &\quad+\frac{(1.439\text{ cm K})(121.1\text{cm}^{-1})}{2+2e^{-(1.439\text{ cm K})(121.1\text{cm}^{-1})/(300\text{ K})}}\frac{(8.31451\text{ J K}^{-1}\text{ mol}^{-1})}{(300\text{ K})}\\ &= \mathbf{11.2}\text{ J K}^{-1}\text{ mol}^{-1}\end{aligned}$$

(c) **(10점)** (2) 식으로부터

$$\begin{aligned}\lim_{T\to\infty}\bar{S}(T) &= \lim_{T\to\infty}\left[R\ln\left(2+2e^{-(1.439\text{ cm K})\bar{\nu}(\text{cm}^{-1})/T}\right)+\frac{(1.439\text{ cm K})\bar{\nu}(\text{cm}^{-1})}{e^{(1.439\text{ cm K})\bar{\nu}(\text{cm}^{-1})/T}+1}\frac{R}{T}\right]\\ &= R\ln 4\end{aligned}$$

2. **(총 10점, 각 5점)**

(a) 수업시간에 배운 이원자분자의 진동 에너지준위 분포 :

$$\begin{aligned}p_n(T) &= \frac{e^{-\beta\epsilon}}{q}\\ &= \frac{e^{-\beta(n+1/2)h\nu}}{e^{-\beta h\nu/2}/(1-e^{-\beta h\nu})}\\ &= e^{-\beta h\nu n}\left(1-e^{-\beta h\nu}\right)\\ &= e^{-n\Theta_{vib}/T}\left(1-e^{-\Theta_{vib}/T}\right)\qquad\left(\Theta_{vib}\equiv\frac{h\nu}{k_B}\right)\end{aligned}\tag{3}$$

$\Theta_{vib}(\text{CO}) = 3084\text{ K}$이므로, (3) 식을 이용하면 25°C에서 각 진동 에너지준위에 있을 분자 수 $N_n(25°\text{C})$는 다음과 같이 구할 수 있다.

$$\begin{aligned}N_0(25°\text{C}) = N_A\times p_0(298.15\text{ K}) &= (6.022\times10^{23})e^{-(0)(3084\text{ K})/(298.15\text{ K})}\left(1-e^{-(3084\text{ K})/(298.15\text{ K})}\right)\\ &= \mathbf{6.022\times10^{23}}\\ N_1(25°\text{C}) = N_A\times p_1(298.15\text{ K}) &= (6.022\times10^{23})e^{-(1)(3084\text{ K})/(298.15\text{ K})}\left(1-e^{-(3084\text{ K})/(298.15\text{ K})}\right)\\ &= \mathbf{1.939\times10^{19}}\end{aligned}$$

(b) (a)와 같은 방식으로, (3) 식을 이용하면 1000°C에서 각 진동 에너지준위에 있을 분자 수 $N_n(1000°\text{C})$는 다음과 같이 구할 수 있다.

$$\begin{aligned} N_0(1000°\text{C}) = N_A \times p_0(1273.15\ \text{K}) &= (6.022\times10^{23})e^{-(0)(3084\ \text{K})/(1273.15\ \text{K})}\left(1-e^{-(3084\ \text{K})/(1273.15\ \text{K})}\right) \\ &= \mathbf{5.488\times10^{23}} \\ N_1(1000°\text{C}) = N_A \times p_1(1273.15\ \text{K}) &= (6.022\times10^{23})e^{-(1)(3084\ \text{K})/(1273.15\ \text{K})}\left(1-e^{-(3084\ \text{K})/(1273.15\ \text{K})}\right) \\ &= \mathbf{4.868\times10^{22}} \end{aligned}$$

3. **(총 20 점, 각 5 점)**

(a)

$$q(V,T) = 1+\gamma e^{-\epsilon/k_BT} \tag{4}$$

(b)

$$Q(N,V,T) = \frac{q^N}{N!} = \frac{(1+\gamma e^{-\epsilon/k_BT})^N}{N!} \tag{5}$$

(c)

$$\begin{aligned} \langle E\rangle &= k_BT^2\left(\frac{\partial \ln Q}{\partial T}\right)_{N,V} \\ &= k_BT^2\frac{\partial}{\partial T}\ln\left[\frac{1}{N!}\left(1+\gamma e^{-\beta\epsilon}\right)^N\right] \\ &= k_BT^2\frac{\partial}{\partial T}\left[N\ln\left(1+\gamma e^{-\epsilon/k_BT}\right)-\ln N!\right] \\ &= Nk_BT^2\frac{\frac{\gamma\epsilon}{k_BT^2}e^{-\epsilon/k_BT}}{1+\gamma e^{-\epsilon/k_BT}} \\ &= \frac{N\gamma\epsilon e^{-\epsilon/k_BT}}{1+\gamma e^{-\epsilon/k_BT}} \end{aligned}$$

(d)

$$
\begin{aligned}
\bar{C}_V &= \left(\frac{\partial\langle E\rangle}{\partial T}\right)_{N_A,V} \\
&= \frac{\partial}{\partial T}\left[\frac{N_A\gamma\epsilon e^{-\epsilon/k_BT}}{1+\gamma e^{-\epsilon/k_BT}}\right] \\
&= N_A\gamma\epsilon\left[\frac{\frac{\epsilon}{k_BT^2}e^{-\epsilon/k_BT}(1+\gamma e^{-\epsilon/k_BT}) - e^{-\epsilon/k_BT}\frac{\gamma\epsilon}{k_BT^2}e^{-\epsilon/k_BT}}{(1+\gamma e^{-\epsilon/k_BT})^2}\right] \\
&= \frac{N_A\gamma\epsilon^2}{k_BT^2}\frac{e^{-\epsilon/k_BT}}{(1+\gamma e^{-\epsilon/k_BT})^2} \\
&= N_Ak_B\gamma\frac{\epsilon^2}{k_B^2T^2}\frac{e^{-\epsilon/k_BT}}{(1+\gamma e^{-\epsilon/k_BT})^2} \\
&= \gamma R\left(\frac{\epsilon}{k_BT}\right)^2\frac{e^{-\epsilon/k_BT}}{(1+\gamma e^{-\epsilon/k_BT})^2} \qquad (6)
\end{aligned}
$$

4. **(총 15점, 각 5점)**

(a) 바닥상태의 에너지는 0이고 첫번째 들뜬상태의 에너지는 300 cm^{-1} 이므로

$$
\begin{aligned}
&p_{e1} = 0.15 = \frac{e^{-\beta\epsilon_{e1}}}{1+e^{-\beta\epsilon_{e1}}} = \frac{e^{-\epsilon_{e1}/k_BT}}{1+e^{-\epsilon_{e1}/k_BT}} = \frac{e^{-(300\ \text{cm}^{-1})hc/k_BT}}{1+e^{-(300\ \text{cm}^{-1})hc/k_BT}} \\
&0.15 + 0.15e^{-(300\ \text{cm}^{-1})hc/k_BT} = e^{-(300\ \text{cm}^{-1})hc/k_BT} \\
&0.15 = 0.85e^{-(300\ \text{cm}^{-1})hc/k_BT} \\
&0.176 = e^{-(300\ \text{cm}^{-1})hc/k_BT} \\
&\ln 0.176 = -\frac{(300\ \text{cm}^{-1})hc}{k_BT} \Rightarrow -1.74 = -\frac{(300\ \text{cm}^{-1})hc}{k_BT} \\
&T = \frac{(300\ \text{cm}^{-1})hc}{1.74\times k_B} = \frac{(300\ \text{cm}^{-1})(6.62608\times10^{-34}\ \text{J s})(2.998\times10^{10}\text{cm s}^{-1})}{(1.74)(1.38066\times10^{-23}\ \text{J K}^{-1})} = 248\ \text{K}
\end{aligned}
$$

(b) 바닥상태의 에너지는 0이고 첫번째 들뜬상태의 에너지는 300 cm^{-1} 이고 들뜬상태의 축퇴도가

2 이므로

$$p_{e1} = 0.15 = \frac{2e^{-\beta\epsilon_{e1}}}{1 + 2e^{-\beta\epsilon_{e1}}} = \frac{2e^{-\epsilon_{e1}/k_BT}}{1 + 2e^{-\epsilon_{e1}/k_BT}} = \frac{2e^{-(300\ \mathrm{cm}^{-1})hc/k_BT}}{1 + 2e^{-(300\ \mathrm{cm}^{-1})hc/k_BT}}$$

$$0.15 + 0.30e^{-(300\ \mathrm{cm}^{-1})hc/k_BT} = 2e^{-(300\ \mathrm{cm}^{-1})hc/k_BT}$$

$$0.15 = 1.70e^{-(300\ \mathrm{cm}^{-1})hc/k_BT}$$

$$0.088 = e^{-(300\ \mathrm{cm}^{-1})hc/k_BT}$$

$$\ln 0.088 = -\frac{(300\ \mathrm{cm}^{-1})hc}{k_BT} \Rightarrow -2.43 = -\frac{(300\ \mathrm{cm}^{-1})hc}{k_BT}$$

$$T = \frac{(300\ \mathrm{cm}^{-1})hc}{1.74 \times k_B} = \frac{(300\ \mathrm{cm}^{-1})(6.62608 \times 10^{-34}\ \mathrm{J\ s})(2.998 \times 10^{10}\mathrm{cm\ s}^{-1})}{(2.43)(1.38066 \times 10^{-23}\ \mathrm{J\ K}^{-1})} = 178\ \mathrm{K}$$

(c) (b) 에서 얻은 온도가 (a) 에서 얻은 온도보다 낮다. 그 이유는 (b) 에는 들뜬 에너지준위에 2 개의 미시상태가 존재하므로 계가 들뜬 에너지준위에 존재할 확률이 증가하기 때문이다.